Prime Radicands Contradict the Printed Density in Conjecture 00000000259

idealistichacker
AI-assisted submission prepared for review

October 22, 2026

Abstract

The printed conjecture concerns primes $p \equiv 1 \pmod 4$, not general composite squarefree radicands. Every such prime admits an integer solution of $x^2 - py^2 = -1$. Consequently the success proportion within that prime class is identically one once the counting denominator is nonzero. The printed constant $1 - \prod_{k\geq 2}(1-2^{-k})$ belongs to $[1/4, 7/16]$ and is not one. We give the descent proof and an actual finite-count/limit formalization. This draft does not claim to resolve a different negative-Pell density problem for composite discriminants, and is not a claim of publication or upstream acceptance.

1 Exact Statement and Interpretation

The English source asserts: "The density of primes $p \equiv 1 \pmod 4$ for which the negative Pell equation is solvable equals $1 - \prod_{k\geq 2}(1 - 2^{-k})$." The Chinese source likewise explicitly specifies primes. Both complete source languages are locked at:

```
Git blob:  15fe4da518462416cfbace31ff72247830e4ea6b
Commit:  b355e42906e6eaf52bc7e9d005e0ef02d8aa7400.
```

Here density within the displayed reference class means the standard relative counting density. Define

$$B_N = \{p < N : p \text{ prime},\ p \equiv 1 \pmod 4\}, \qquad A_N = \{p \in B_N : \exists x, y \in \mathbb{Z},\ x^2 - py^2 = -1\}.$$

The assertion formalized and refuted is

$$\frac{|A_N|}{|B_N|} \longrightarrow C := 1 - \prod_{k\geq 2}(1 - 2^{-k}) \quad (N \longrightarrow \infty).$$

The denominator is explicitly disclosed; the Lean theorem is not advertised as a proof about a different density denominator. No restriction to primes with an additional class-number condition is inserted. A prime $p \equiv 1 \pmod 4$ is already a squarefree positive fundamental discriminant. The introductory wide-class-number assertion is not needed as an assumption: one false necessary density clause suffices to refute the displayed conjunction under this reading.

2 All Eligible Primes Give Negative Pell Solutions

Theorem 1. *For every prime $p \equiv 1 \pmod 4$, there exist $u, v \in \mathbb{Z}$ with $u^2 - pv^2 = -1$.*

Proof. The ordinary Pell existence theorem supplies an integer solution $x^2 - py^2 = 1$ whose $x > 1$ is minimal among positive Pell solutions with $x > 1$. This is applicable because a positive prime is not a square. Reducing modulo four shows x odd and y even. Set

$$a = (x-1)/2 > 0, \qquad b = (x+1)/2 = a + 1, \qquad c = y/2.$$

Then a, b are coprime positive integers and $ab = pc^2$. Unique factorization and coprimality put the prime p in exactly one factor. Each remaining prime exponent is even, so either

$$a = pu^2, \quad b = v^2, \qquad \text{or} \qquad a = u^2, \quad b = pv^2.$$

In the first case $v^2 - pu^2 = b - a = 1$. Since $b = a + 1 > 1$ and $v^2 = (x+1)/2$, we have $1 < |v| < x$. Thus $(|v|, |u|)$ is a positive Pell solution with smaller first coordinate, contradicting minimality. The second case therefore holds, giving $u^2 - pv^2 = a - b = -1$. □

The Lean proof invokes Mathlib's actual `Pell.IsFundamental.exists_of_not_isSquare`; it does not assume the required negative-Pell solution. Its descent uses actual integer Pell solutions and a proved split of adjacent coprime square factors. As a diagnostic example, $p = 109$ has the exact certificate

$$8890182^2 - 109 \cdot 851525^2 = -1.$$

A bounded search that stops before $y = 851525$ is not a proof of insolubility. The general argument above, not this isolated certificate, proves the theorem.

3 The Printed Constant Is Separated From One

Write $f_i = 1 - 2^{-(i+2)}$ for $i \in \mathbb{N}$ and $P_n = \prod_{i<n} f_i$. The geometric series of deviations is summable, so $\prod_i f_i$ exists. All factors lie in $[0, 1]$.

For $0 \le a_i \le 1$, the elementary union bound for products states

$$\prod_{i<n} (1 - a_i) \ge 1 - \sum_{i<n} a_i.$$

It follows by induction: if $Q \le 1$ and $Q \ge 1 - S$, then $Q(1-a) = Q - aQ \ge Q - a \ge 1 - S - a$. The shifted geometric tail satisfies $\sum_{i<n} 2^{-(i+3)} \le 1/4$. Splitting off the first factor gives

$$\frac{9}{16} \le P_{n+1} = \frac{3}{4} \prod_{i<n} (1 - 2^{-(i+3)}) \le \frac{3}{4}.$$

Passing to the genuine infinite-product limit yields

$$\frac{9}{16} \le \prod_{k \ge 2} (1 - 2^{-k}) \le \frac{3}{4}, \qquad \frac{1}{4} \le C \le \frac{7}{16} < 1.$$

These are exact rational inequalities, not numerical approximations to an uncomputed product. Lean proves summability, multipliability, finite-product bounds, and the passage to the limit.

4 Counting and Closed Limit Negation

Theorem 2. *$A_N = B_N$ for every $N \in \mathbb{N}$. For every $N \geq 6$, $|A_N|/|B_N| = 1$. In particular this ratio cannot tend to C.*

Proof. The prime-radicand theorem applies to each member of B_N, so $A_N = B_N$. For $N \geq 6$, $5 \in B_N$, making $|B_N| > 0$. Dividing equal counts gives one. The ratio is therefore eventually the constant one and converges to one. Uniqueness of real limits and $C \neq 1$ negate convergence to the conjectured value. No prime number theorem or statistical heuristic is required for this relative-count identity. □

5 Formalization and Verification Boundary

The research sources pin Lean 4.33.1. The Mathlib commit is:

`0df444a360eaa60ab8c11dca51a86af692955474.`

- `PrimePell.lean`: four general integer Pell theorems.
- `ProductBounds.lean`: eight actual infinite-product and bound theorems.
- `RelativeDensity.lean`: seven count, limit, negation, and certificate theorems, with the actual sets A_N, B_N defined using finite-set filters.
- `Main.lean`: imports the complete mathematical argument.
- `Check.lean`: all 19 actual theorem types and axiom audits.

The decisive declaration is

```
PellDensity259.not_claimed_relative_density :
  not (Filter.Tendsto relativeDensity Filter.atTop
    (nhds PellConstant259.claimedConstant)).
```

It has no external density, solution-existence, or product-separation premise. Local warning-as-error source builds and all axiom audits have succeeded; only the standard foundations `propext`, `Classical.choice`, and `Quot.sound` occur. The package reproducer verifies all nine physical pinned Git dependencies, their origins, tracked source bytes, and index/tree agreement before and after execution. It cleans only this project's build outputs, rebuilds Main, replays every mathematical source and Main with warnings as errors, and audits all 19 actual types and axiom footprints. Its exact integer certificates and finite product checks are regression diagnostics, not replacements for the Lean proofs. Installed Lean and dependency build caches remain trust inputs: this is not an empty-machine bootstrap or a clean build of all Mathlib dependencies. Independent review is internal AI-assisted review, not human, institutional, or organizer endorsement. Submission requests upstream review; it is not a claim of acceptance or first-solver credit.